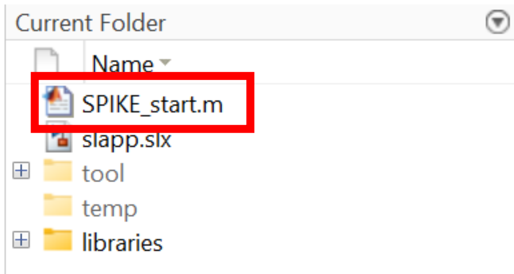
How to use SPIKE Library

1. How to start

a. Run SPIKE_start.m to add the required paths.



b. Open slapp.slx. This launches the SPIKE Simulink model.

Note:
You can rename the Simulink model if desired.

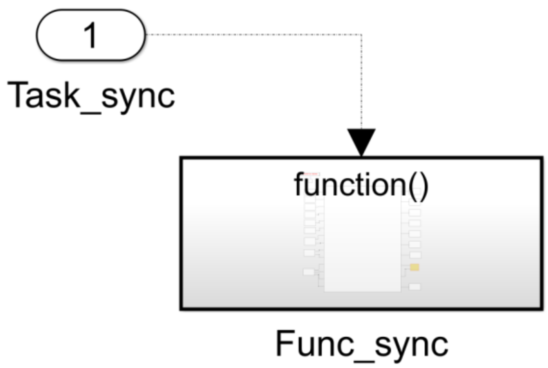
2. Creating a model

The model consists of three layers:

A. Task layer
B. Interface layer
C. Application layer.

A. Task layer

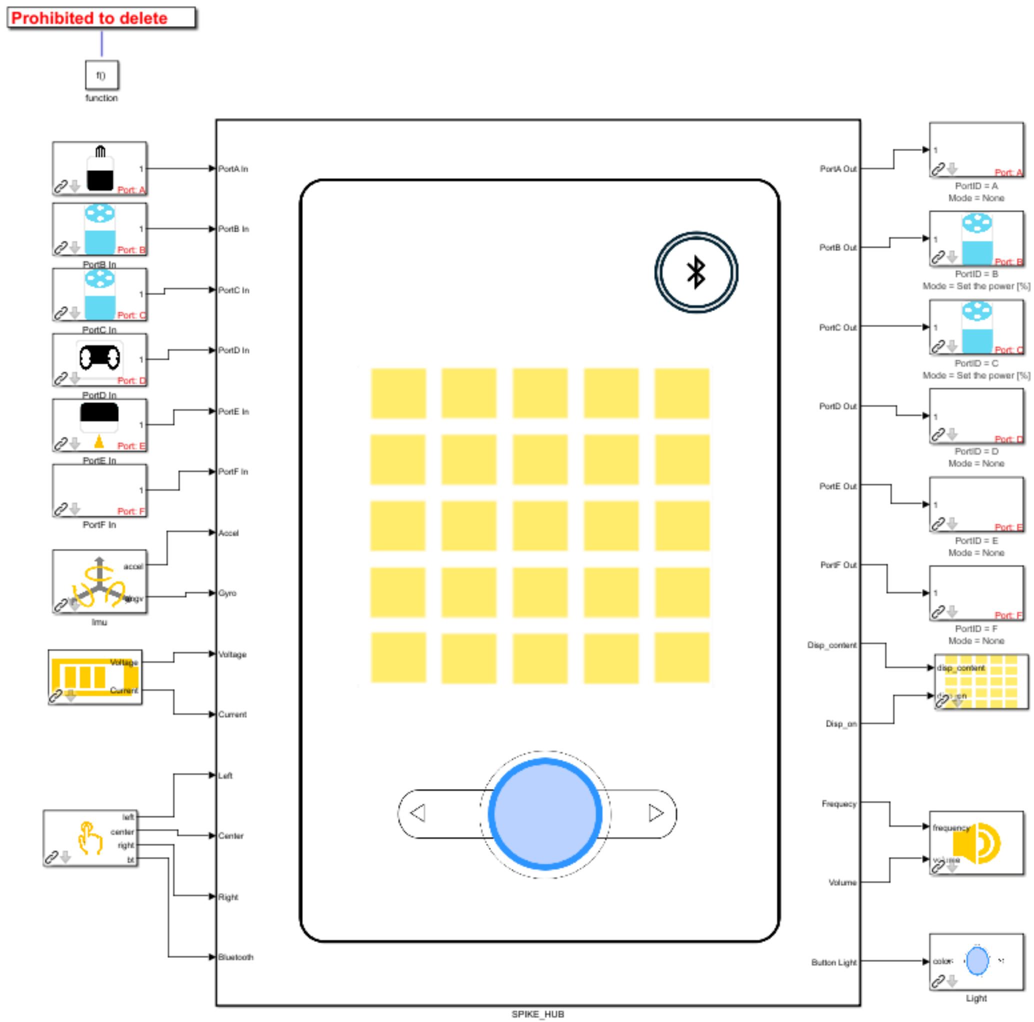
Please do not change this top task layer



The task layer is located at the top of the model. Do not modify this layer.

The task is implemented using a Function-Call Subsystem, and the internal logic executes at a fixed cycle of 4 ms.

B. Interface layer



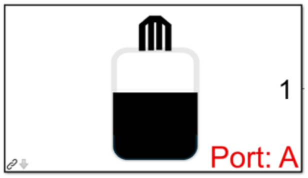
The interface layer is located inside the Func_sync subsystem of the task layer. It defines the interface to the sensors and actuators connected to the SPIKE Prime hub.

- Sensor inputs are shown on the left-hand side.
- Actuators and display-related blocks are grouped on the right-hand side.

Open the block parameters of each block to see usage instructions.
If different sensors or actuators are connected to a port, update the corresponding block settings. If nothing is connected, set the port to **None**.

【Sensor block specification】

1) **Force sensor**
   Get the load value.

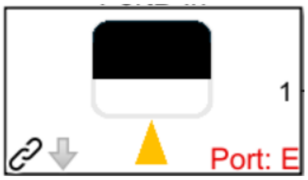


   Output signal type: double
   Output signal unit: N
   Output signal range:0 - about 10

   LEGO specification document:
   https://assets.education.lego.com/v3/assets/blt293eea581807678a/blt23df304b05e587b2/5f8801ba721f8178f2e5e626/techspecs_technicforcesensor.pdf?locale=en-us

2) **Color sensor**

Get the value of the reflection, ambient light, or color (HSV).



- **Reflection mode**
  Get the percentage of how much a surface reflects the light emitted by the sensor.
  Output signal type: double
  Output signal unit: %
  Output signal range:0-100 (100= white, 50= gray, 01=black, 00=No surface)

- **Ambient mode**
  Get the ambient light intensity.
  Output signal type: double
  Output signal unit: %
  Output signal range:0-100 (0=dark, 100=bright)

- **Color mode**
  Get the color of surface.
  Output signal type: integer
  Output signal unit: Non-dimensional
  Output signal range: 0-7 (0=Others, 1=Red, 2=Orange, 3=Yellow, 4=Green, 5=Blue, 6=Indigo, 7=Violet)

LEGO specification document:
https://assets.education.lego.com/v3/assets/blt293eea581807678a/blt62a78c227edef070/5f8801b9a302dc0d859a732b/techspecs_techniccolorsensor.pdf?locale=en-us

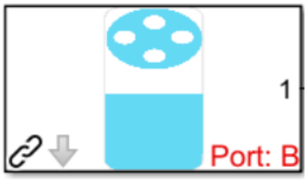
3) **Ultrasonic sensor**

Get the distance in front of the ultrasonic sensor.

Output signal type: double
Output signal unit: mm
Output signal range: 0-2000

LEGO specification document:
https://assets.education.lego.com/v3/assets/blt293eea581807678a/blt64c2b9534cf10f68/5f8801b8bc43790f5c4389ea/techspecs_technicdistancesensor.pdf?locale=en-us

4) **Rotation sensor**
Get the angle and angle velocity of the motor. The angle counts 360° per revolution. Angular velocity is calculated internally from the angle.



- **Angle**
  Output signal type: double
  Output signal unit: deg
  Output signal range: Nothing

- **Angular velocity**
  Output signal type: double
  Output signal unit: deg/s
  Output signal range: Nothing

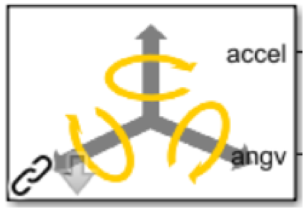
LEGO specification document:

- Medium motor
  https://assets.education.lego.com/v3/assets/blt293eea581807678a/blt692436dd1e8fa71c/5f8801d5c8a27c1d9614c27e/techspecs_technicmediumangularmotor.pdf?locale=en-us

- Large motor
  https://assets.education.lego.com/v3/assets/blt293eea581807678a/bltb9abb42596a7f1b3/5f8801b5f4c5ce0e93db1587/le_spike-prime_tech-fact-sheet_45602_1hy19.pdf?locale=en-us

5) IMU

Get the acceleration and angular velocity.

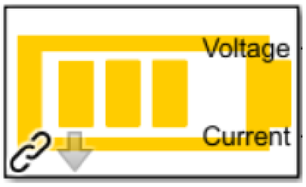


- **Acceleration**
  This is a vector signal which contains x, y, z axes of the acceleration.
  Output signal type: double
  Output signal unit: mm/s^2
  Output signal range: Nothing

- **Angular velocity**
  This is a vector signal which contains roll, pitch and yaw of the angular velocity.
  Output signal type: double
  Output signal unit: deg/s
  Output signal range: Nothing

LEGO specification document:
https://assets.education.lego.com/v3/assets/blt293eea581807678a/bltf512a371e82f6420/5f8801baf4f4cf0fa39d2feb/techspecs_techniclargehub.pdf?locale=en-us

6) **Battery**
Retrieve the battery voltage and the battery current.



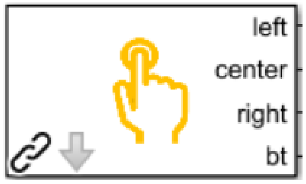
- **Voltage**
  Output signal type: double
  Output signal unit: mV
  Output signal range: Nothing

- **Current**
  Output signal type: double
  Output signal unit: mA
  Output signal range: Nothing

  LEGO specification document:
  https://assets.education.lego.com/v3/assets/blt293eea581807678a/blt706bcb108ec66a2b/6630d4aeba17b0181acb6d3d/SPIKE_Prime_Battery_Tech_Spec_Sheet.pdf?locale=en-us

7) **Buttons**

Checks which buttons (left, center, right, bluetooth) are currently pressed.



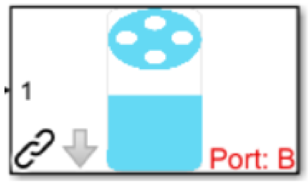
Output signal type: double
Output signal unit: Non-dimensional
Output signal range: 0(Not pressed) or 1(Pressed)

LEGO specification document:
https://assets.education.lego.com/v3/assets/blt293eea581807678a/bltf512a371e82f6420/5f8801baf4f4cf0fa39d2feb/techspecs_techniclargehub.pdf?locale=en-us

【Actuator block specification】

1) **Motor**

Control the motor according to the settings and the input.



- **Motor mode**
  a) Set the speed of rotation
  Control the motor speed at the specified value. Control method and parameters cannot be changed.
  Input signal type: double
  Input signal unit: deg/s
  Input signal range: Nothing

  b) Set the power of the motor
  Set the motor voltage.
  Input signal type: double
  Input signal unit: %
  Input signal range: -100~100[%]

  c) Stop
  Stop the motor. But the encoder is active.
  Input signal type: any signal is ok

  d) None
  Stop the motor. And the encoder is also inactive.
  Input signal type: any signal is ok

- **Encoder mode**
  a) No reset
  Initial encoder value is set to an absolute angle.

  b) Reset
  Initial encoder value is set to 0.

- Direction
  Defines whether clockwise or anti-clockwise rotation is treated as positive.
    a) Clockwise
    b) Couter-clockwise

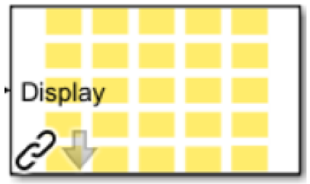
LEGO specification document:

- Medium motor
https://assets.education.lego.com/v3/assets/blt293eea581807678a/blt692436dd1e8fa71c/5f8801d5c8a27c1d9614c27e/techspecs_technicmediumangularmotor.pdf?locale=en-us

- Large motor
https://assets.education.lego.com/v3/assets/blt293eea581807678a/bltb9abb42596a7f1b3/5f8801b5f4c5ce0e93db1587/le_spike-prime_tech-fact-sheet_45602_1hy19.pdf?locale=en-us

2) **Display**

Light up a 5x5 matrix display. There are three ways to specify how to make it glow, and the display direction can be specified as up, down, left or right.



- **Display mode**

    a) Numeric

    Displays a number. A minus sign (-) is shown as a faint dot in the center of the display.
    Input signal type: double
    Input signal unit: Non-dimensional
    Input signal range: -99 to 99

    b) Character

    Display one character at a time.
    Input signal type: string
    Input signal unit: Non-dimensional
    Input signal range: letter (a–z), capital letter (A–Z) or
    symbols: !"#$%&'()*+,-./:;<=>?@[]^_`{|}

    c) Matrix

    Light up any pattern you like. Each element of the vector corresponds to a respective LED and specifies the luminance as a number from 0-100.
    Input signal type: integer array (size 25×1)
    Each element of the vector corresponds to each LED
    Input signal unit: Non-dimensional
    Input signal range: 0-100
    Specifies the luminance as a number from 0-100.
    (0=dark, 100=bright)

LEGO specification document:
https://assets.education.lego.com/v3/assets/blt293eea581807678a/bltf512a371e82f6420/5f8801baf4f4cf0fa39d2feb/techspecs_techniclargehub.pdf?locale=en-us

3) **Speaker**

Tone sound with the speaker.

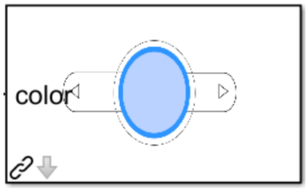
- **frequency**
  Input signal type: double
  Input signal unit: Hz
  Input signal range: 0-24000
  0: silent, other: sound the corresponding tone

- **volume**
  Input signal type: double
  Input signal unit: %
  Input signal range: 0-100

LEGO specification document:
https://assets.education.lego.com/v3/assets/blt293eea581807678a/bltf512a371e82f6420/5f8801baf4f4cf0fa39d2feb/techspecs_techniclargehub.pdf?locale=en-us

4) **Light**

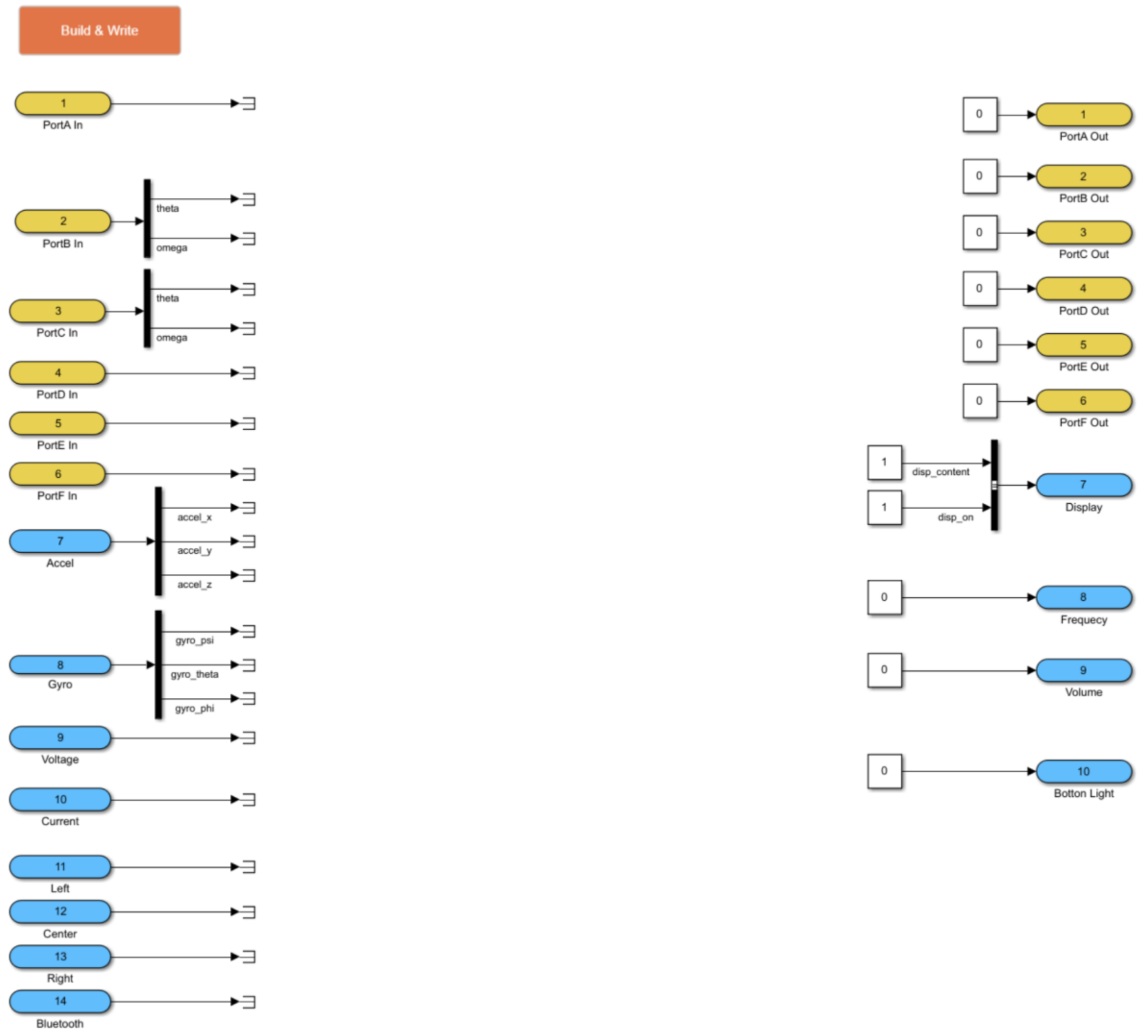
Turn the light on with specified color.



Input signal type: double
Input signal unit: Non-dimensional
Input signal range: Nothing
0=White, 1=Red, 2=Orange, 3=Yellow, 4=Green, 5=Blue,
6=Indigo, 7=Purple, Others=Off

LEGO specification document:
https://assets.education.lego.com/v3/assets/blt293eea581807678a/bltf512a371e82f6420/5f8801baf4f4cf0fa39d2feb/techspecs_techniclargehub.pdf?locale=en-us

C. Application layer



Implement control logic in this layer.

- Only discrete systems are supported.
- Block sample time must be set to **−1**.

To change the control execution period (default: 4 ms), do not modify block sample times. Instead, edit slapp.cfg in: libraries\libspikert

[Note]

- Sensor and actuator blocks do not operate during normal simulation. Use a test harness for simulation.
- Some blocks (e.g., 1D Lookup Table) require external C libraries and may not be supported.
- Model Reference is not supported.
- Always create models using slapp.slx as a template. Creating a blank model is not supported. Copying or renaming slapp.slx is allowed.

3. Building & writing a model

After completing the application layer, deploy the model to the SPIKE Prime hub.

A. Connecting SPIKE Prime to a computer

1. Power off the SPIKE Prime hub.
2. Press and hold the Bluetooth button while connecting the USB cable to the PC.
3. When connected, the Bluetooth button lights up purple.
4. Continue holding the Bluetooth button until the LED cycles: red → green → blue → off → red → …, then release.

B. Build & Write

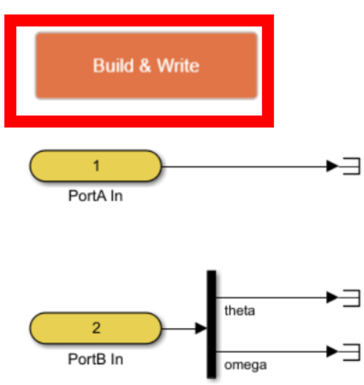
Click ‘Build & Write**’** in the top-left corner of the Application layer.
This process:

- Generates code
- Builds the executable
- Writes it to the hub

The program starts automatically after writing complete.
After startup:

- The center LED swirls
- The center button flashes blue

Disconnect the USB cable once confirmed. Build and write logs are displayed in the MATLAB Command Window. If writing fails, review the error messages and take corrective action.



4. Restoring the Original Firmware

After writing custom code, the original LEGO SPIKE firmware is replaced, and the standard LEGO SPIKE App can no longer be used.

Restore the original firmware using the following link:

https://spikelegacy.legoeducation.com/hubdowngrade/#step-1